

Advanced (Habanero)

- Q1.** What is the remainder when 24 is divided by 5?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 0
- Q2.** Which number is divisible by 3 and 5?
(A) 10
(B) 15
(C) 20
(D) 25
(E) 30
- Q3.** If a is divisible by 2 and b is divisible by 3, then $a \cdot b$ is divisible by?
(A) 2
(B) 3
(C) 5
(D) 6
(E) 9
- Q4.** What is the result of $17 \div 5$ with remainder?
(A) $3R2$
(B) $3R1$
(C) $3R3$
(D) $3R4$
(E) $3R0$
- Q5.** A number is divisible by 9 if the sum of its digits is?
(A) even
(B) odd
(C) divisible by 3
(D) divisible by 9
(E) a prime number
- Q6.** What is $27 \div 3$ using long division?
(A) 8
(B) 9
(C) 10
(D) 11
(E) 12
- Q7.** If a number is divisible by 10, it must also be divisible by?
(A) 2 and 3
(B) 2 and 5
(C) 3 and 5
(D) 2 and 9
(E) 5 and 9
- Q8.** What is $48 \div 6$ with remainder?
(A) $7R5$
(B) $8R0$
(C) $6R4$
(D) $8R1$
(E) $8R2$

Open Ended Questions (FRQ)

Q9. Perform the long division to find $63 \div 9$. Express the result with remainder.

Q10. A bookshelf has 48 books. If the books are packed in boxes that hold 8 books each, how many boxes are needed and how many books will be left over?

Chef's Tips

- Emphasize the importance of using divisibility rules to simplify calculations.
- Encourage students to use long division to find quotients and remainders.
- Remind students to express division results with remainders.